This exam has 5 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

By participating in this exam, you underwrite the following `Statement of Integrity`:

I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to make this exam completely on my own, only consult those sources that are allowed explicitly, not share my solutions with other students, and make myself available for any oral clarifications regarding this exam.

1. Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

Solution: The Sheffer stroke is not associative. The (in total four) counterexamples to the associativity law for the Sheffer stroke, $(\phi|\psi)|\chi \equiv \phi|(\psi|\chi)$, are if ϕ and χ have opposite truth values.

If $\phi = T$ and $\chi = F$, then $(\phi|\psi)|\chi \equiv (\phi|\psi)|F \equiv T$ while $\phi|(\psi|\chi) \equiv T|T \equiv F$.

If $\phi = F$ and $\chi = T$, then $(\phi|\psi)|\chi \equiv T|T \equiv F$ while $\phi|(\psi|\chi) \equiv F|(\psi|\chi) \equiv T$.

2. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF

$$(\neg p \vee q \vee \neg r) \ \wedge \ (\neg p \vee \neg q \vee \neg r) \ \wedge \ (p \vee \neg q \vee \neg r) \ \wedge \ r$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p, then on q, and then on r; and let it always try the truth value true ($\top$) before false ($\bot$).

Solution: The conjunct r is a unit clause, so substitute $\top$ for it:

$$(\neg p \vee q \vee \neg\top) \ \wedge \ (\neg p \vee \neg q \vee \neg\top) \ \wedge \ (p \vee \neg q \vee \neg\top) \ \wedge \ \top$$

After replacing the two occurrences of $\neg\top$ by $\bot$, this reduces to

$$(\neg p \vee q) \ \wedge \ (\neg p \vee \neg q) \ \wedge \ (p \vee \neg q)$$

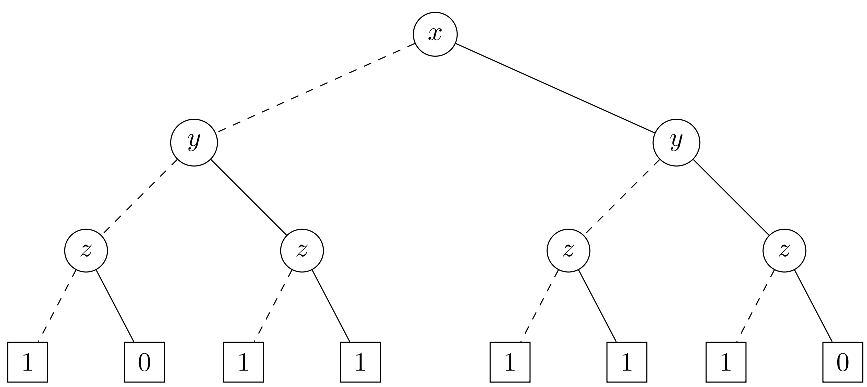
This CNF formula contains no occurrences of unit clauses or pure variables.

Substituting $\top$ for p, replacing the two occurrences of $\neg\top$ by $\perp$, and reducing the formula yields $q \wedge \neg q \wedge \top$, which reduces further to $q \wedge \neg q$. After substituting $\top$ for q the formula reduces to $\perp$. Now backtrack and substitute $\perp$ for q. Again the formula reduces to $\perp$.

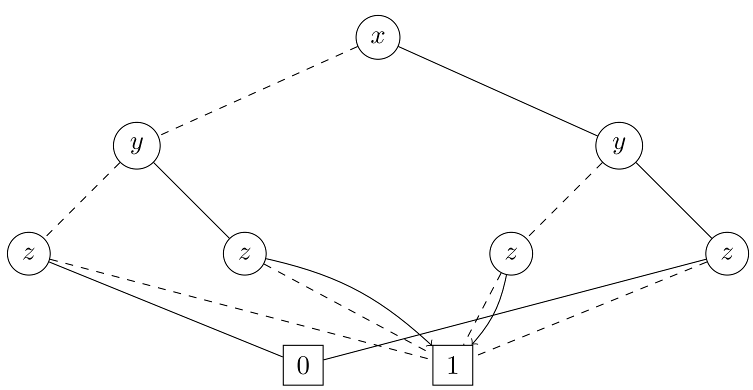
Now backtrack and substitute $\perp$ (instead of $\top$) for p. Replacing the two occurrences of $\neg\perp$ by $\top$ and reducing the formula yields $\top \wedge \top \wedge \neg q$, which reduces further to $\neg q$. After substituting $\top$ for q, the formula reduces to $\perp$. Now backtrack and substitute $\perp$ for q, in which case the formula reduces to $\top$.

3. OBDD (12 points)
Represent $z \to (x \oplus y)$ by means of a binary decision tree, with respect to the variable ordering x, y, z. Moreover, reduce this binary decision tree to an ordered binary decision diagram. (Also give the intermediate reduction steps.)

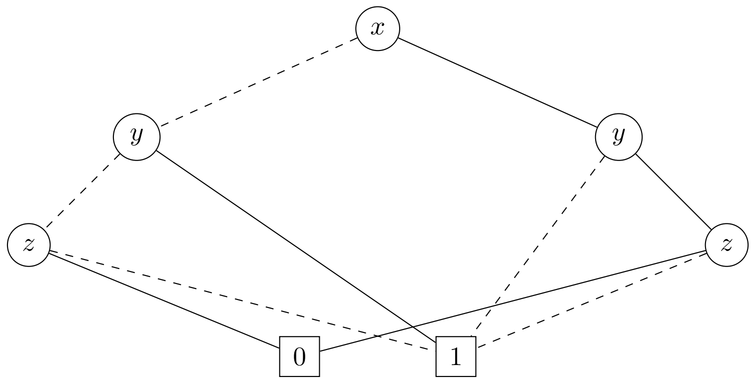
Solution: The formula is true if z is false or $x \oplus y$ is true. This can be expressed by the following binary decision tree.



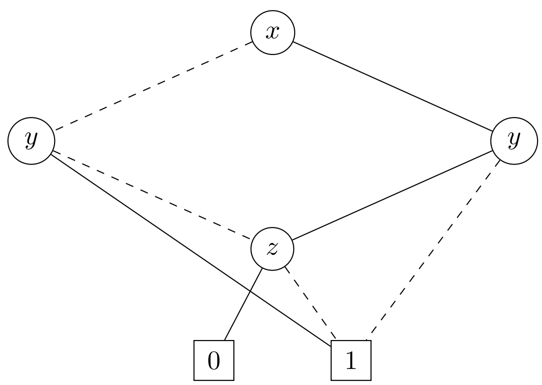
Apply C1 to collapse the 0- and the 1-leaves.



Apply C2 twice to remove the second and the third z node.



Finally, apply C3 to collapse the two remaining z nodes. This results in the reduced OBBD.



4. Predicate logic (11 points)
Give two models to show that the formulas $C(x) \to D(x)$ and $D(x) \to C(x)$ do not semantically entail each other. (Explain your answer.)

Solution: As a first model, let $\mathcal{M}_1 = \{a\}$ where $C(a)$ does not hold while $D(a)$ does. If x is interpreted as a, then $C(x) \to D(x)$ holds in $\mathcal{M}_1$ while $D(x) \to C(x)$ does not. So the first formula does not semantically entail the second formula.

As a second model, let $\mathcal{M}_2 = \{b\}$ where $C(b)$ holds while $D(b)$ does not. If x is interpreted as b, $C(x) \to D(x)$ does not hold in $\mathcal{M}_2$ while $D(x) \to C(x)$ does. So the second formula does not semantically entail the first formula.

5. Sets (6+6 points)
Consider the following set-theoretic equality

$$(((A' \cap C) \cup C') \cap B)' \;=\; B' \cup (A \cap C).$$

(a) Prove it using the rules of the algebra of sets.

Solution:

$$\begin{array}{rcl}
(((A' \cap C) \cup C') \cap B)' & = & \text{(De Morgan)} \\
((A' \cap C) \cup C')' \cup B' & = & \text{(De Morgan)} \\
((A' \cap C)' \cap C'') \cup B' & = & \text{(Involution)} \\
((A' \cap C)' \cap C) \cup B' & = & \text{(De Morgan)} \\
((A'' \cup C') \cap C) \cup B' & = & \text{(Involution)} \\
((A \cup C') \cap C) \cup B' & = & \text{(Distributivity)} \\
((A \cap C) \cup (C' \cap C)) \cup B' & = & \text{(Complement)} \\
((A \cap C) \cup \emptyset) \cup B' & = & \text{(Identity)} \\
(A \cap C) \cup B' & = & \text{(Reorder)} \\
B' \cup (A \cap C). & &
\end{array}$$

(b) Assuming that $\#U = 14$, $\#A = 5$, $\#B = 7$, $\#C = 6$, $\#(A \cup C) = 9$, and $\#(A \cap B' \cap C) = 1$, compute the number of elements in $B' \cup (A \cap C)$.
Solution:

$$\#(B' \cup (A \cap C)) = \#B' + \#(A \cap C) - \#(A \cap B' \cap C) =$$
$$(\#U - \#B) + (\#A + \#C - \#(A \cup C)) - \#(A \cap B' \cap C) =$$
$$14 - 7 + 5 + 6 - 9 - 1 = 8.$$

6. Relations (3+4+4 points)
In the set $V = \{a, b, c, d\}$ consider the relations

$$\begin{array}{rcl}
R & = & \{\langle a,a\rangle, \langle b,a\rangle, \langle b,b\rangle, \langle b,d\rangle, \langle c,c\rangle, \langle d,c\rangle, \langle d,d\rangle\}, \\
S & = & \{\langle a,b\rangle, \langle d,c\rangle\}.
\end{array}$$

(a) Write down the matrix representation of the relation R.

Solution: $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

(b) Is the relation R reflexive? Is it transitive? Is it an ordering relation?
Solution: Yes, R is reflexive, since $\langle a,a\rangle, \langle b,b\rangle, \langle c,c\rangle, \langle d,d\rangle \in R$;
No, R is not transitive, since e.g. $\langle b,d\rangle, \langle d,c\rangle \in R$, $\langle b,c\rangle \notin R$;
No, R is not an ordering relation, since e.g. it is not transitive.

(c) Determine the relation $S \circ R$ by listing all its elements in the curly-bracket notation.
Solution: $S \circ R = \{\langle a,b\rangle, \langle b,b\rangle, \langle b,c\rangle, \langle d,c\rangle\}$

7. Functions (5+6 points)
We are given the functions $sqr : \mathbb{R} \to \mathbb{R}$, $mul_3 : \mathbb{R} \to \mathbb{R}$, $sub_2 : \mathbb{R} \to \mathbb{R}$ defined by

$$sqr(x) \;=\; x^2, \qquad mul_3(x) \;=\; 3x, \qquad sub_2(x) \;=\; x - 2.$$

(a) For each function, determine whether or not the function is injective.
Solution: No, sqr is not injective, since e.g. $sqr(-1) = sqr(+1)$;
Yes, mul_3 is injective, since $3x = 3y$ gives $x = y$;
Yes, sub_2 is injective, since $x - 2 = y - 2$ gives $x = y$.

(b) Express the function $g : \mathbb{R} \to \mathbb{R}$ defined by

$$g(x) \;=\; \frac{1}{3}x^2 + \frac{2}{3}$$

as a composition of sqr, mul_3, sub_2, and their inverses.
Solution: $g(x) = \dfrac{1}{3}x^2 + \dfrac{2}{3} = \dfrac{1}{3}(x^2 + 2) = (mul_3^{-1} \circ sub_2^{-1} \circ sqr)(x)$.

8. Induction and recursion (3+8 points)
Consider the sequence $(t_n)_{n\in\mathbb{N}}$ of numbers defined recursively by

$$t_0 \;:=\; \frac{1}{2}, \qquad t_{n+1} \;:=\; t_n + 3^n + \frac{1}{2}.$$

We claim that the following statement is true for all natural numbers n:

$$t_n \;=\; \frac{1}{2}\left(3^n + n\right).$$

(a) Verify by explicit computation that the claim is true for $n = 0$, $n = 1$ and $n = 2$.
Solution: $t_0 = \frac{1}{2} = \frac{1}{2}(3^0 + 0)$;
$t_1 = \frac{1}{2} + 3^0 + \frac{1}{2} = 2 = \frac{1}{2}(3^1 + 1)$;
$t_2 = 2 + 3^1 + \frac{1}{2} = 5\frac{1}{2} = \frac{1}{2}(3^2 + 2)$.

(b) Prove by mathematical induction that the statement holds true for all natural numbers n.
Solution:

- *Base case:* $t_0 = \frac{1}{2} = \frac{1}{2}(3^0 + 0)$.
- *Induction step:* Let $m \geq 0$. Assume $t_m = \frac{1}{2}(3^m + m)$ (IH).

  Then $t_{m+1} = t_m + 3^m + \frac{1}{2} \overset{\text{(IH)}}{=} \frac{1}{2}(3^m + m) + 3^m + \frac{1}{2} = \frac{3}{2} \cdot 3^m + \frac{1}{2}m + \frac{1}{2}$ and $\frac{1}{2}(3^{m+1} + (m+1)) = \frac{3}{2} \cdot 3^m + \frac{1}{2}m + \frac{1}{2}$.
  Hence (IH) implies $t_{m+1} = \frac{1}{2}(3^{m+1} + (m+1))$.

After completing the exam, show your solutions to the camera before closing Proctorio.

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